



CHAPTER 04



Lists And Tuples

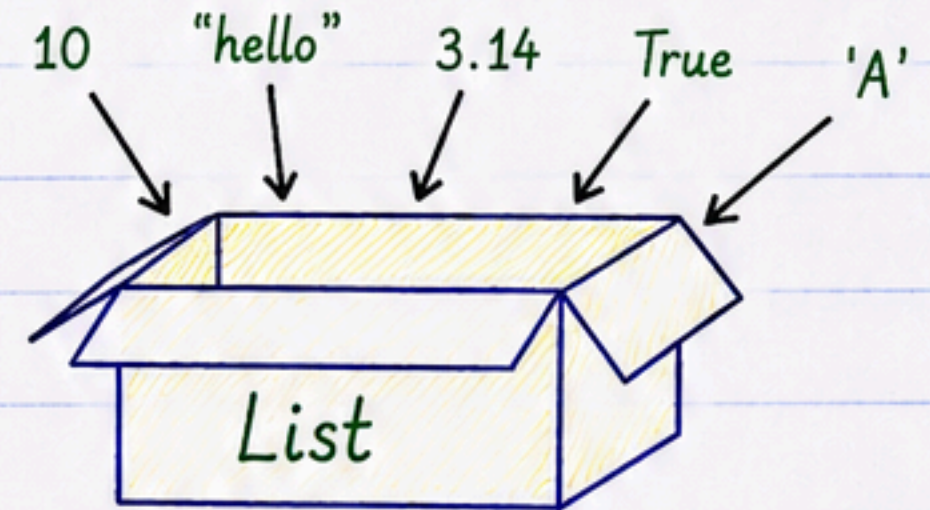
Python **lists** are containers to store a set of **values** of any data type.

Example:

```
my_list = [10, "hello", 3.14, True, 'A']
```

Here, the list contains an integer, string, float, boolean and character.

A list can store different types of data together!



1. List Indexing

A list can be **indexed** just like a string.

```
l1 = [7, 9, "harry"]
```

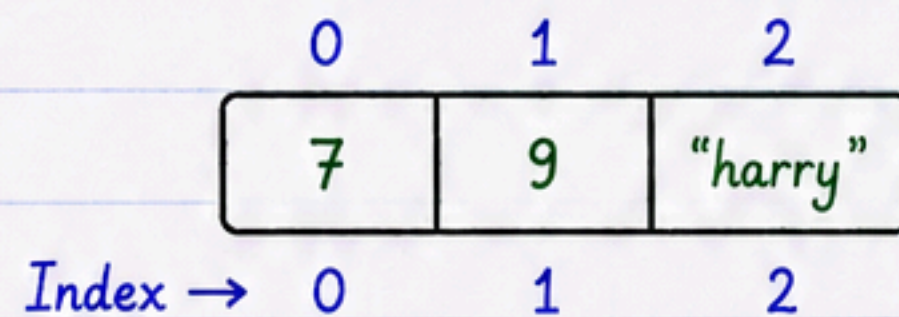
```
l1 = [7, 9, "harry"]
```

```
l1[0] # 7
```

```
l1[1] # 9
```

```
l1[2] # harry
```

```
l1[0:2] # [7, 9] # list slicing
```



```
l1[0:2] → [7, 9] ----
```

(start index included, end index excluded)



2. List Methods

Consider the following list.

```
l1 = [1, 8, 7, 2, 21, 15]
```



• **l1.sort():** updates the list to → [1, 2, 7, 8, 15, 21]

• **l1.reverse():** updates the list to → [15, 21, 2, 7, 8, 1]

• **l1.append(8):** adds 8 at the end of the list → [1, 8, 7, 2, 21, 15, 8]

• **l1.insert(3, 8):** This will add 8 at 3 index → [1, 8, 7, 8, 2, 21, 15]



Practice makes perfect!





- `l1.pop(2)`: Will delete element at index 2 and return its value.

`l1 = [1, 8, 7, 2, 21, 15]`

1	8	7	2	21	15
0	1	2	3	4	5

→ `l1 = [1, 8, 2, 21, 15]`

element at index 2 deleted

7
returned

- `l1.remove(21)`: Will remove 21 from the list.

`l1 = [1, 8, 7, 2, 21, 15]`

1	8	7	2	21	15
---	---	---	---	----	----

→ `l1 = [1, 8, 7, 2, 15]`

21 removed from the list



★ Tuples In Python

A tuple is an **immutable** data type in python.

Immutable means it cannot be changed after creation.



```
a = ()      # empty tuple
a = (1,)    # tuple with only one element
            # needs a comma
a = (1, 7, 2) # tuple with more than
            # one element
```

Tuple

(1 , 7 , 2)

Index → 0 1 2



Tuple Methods

Consider the following tuple.

`a = (1, 7, 2)`

`a = (1, 7, 2)`

1	7	2
0	1	2



- `a.count(1)`: a count (1) will return number of times 1 occurs in a.

`a = (1, 7, 2)`

1	7	2
0	1	2

→ 1 (occurs 1 time)

- `a.index(1)`: will return the index of first occurrence of 1 in a.

`a = (1, 7, 2)`

1	7	2
0	1	2

→ 0 (index of first occurrence of 1)



Keep practising!



Lists are mutable, Tuples are immutable!

Practice Set

Practice makes Perfect!

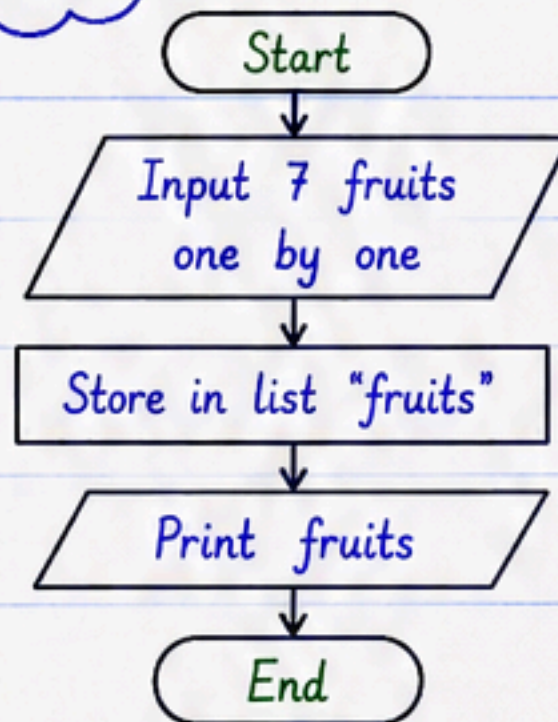


1. Write a program to store seven fruits in a list entered by the user.



Logic / Steps :

1. Take input of 7 fruits from the user.
2. Store them in a list.
3. Display the list.



Example Output

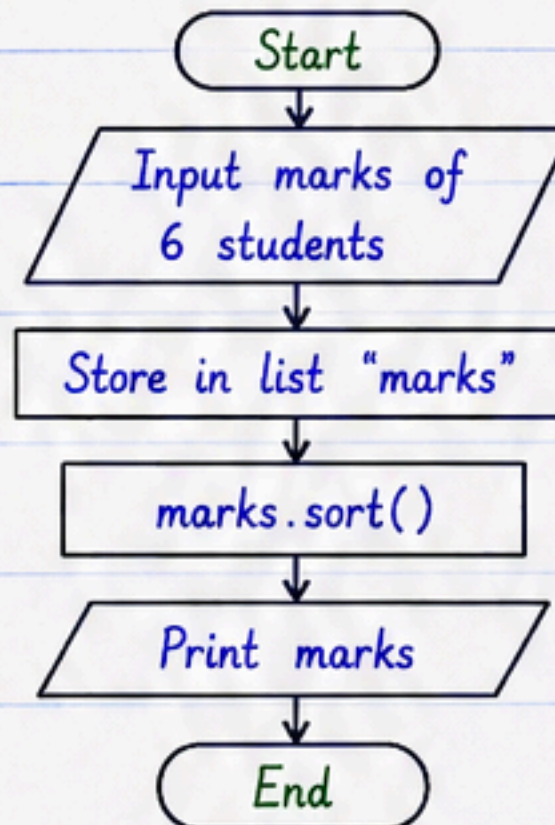
Fruits list:
["apple", "banana", "mango", "orange", "grapes", "guava", "papaya"]

2. Write a program to accept marks of 6 students and display them in a sorted manner.



Logic / Steps :

1. Take input of marks of 6 students.
2. Store them in a list.
3. Sort the list.
4. Display the sorted list.



Example Output

Sorted Marks:
[45, 56, 67, 78, 89, 95]

.sort() method arranges in ascending order

3. Check that a tuple type cannot be changed in python.



Logic / Steps :

1. Create a tuple.
2. Try to change an element.
3. Observe the error.

Example :

```
t = (1, 2, 3, 4)
t[1] = 10 # This will give an error
TypeError: 'tuple' object does not support item assignment
```

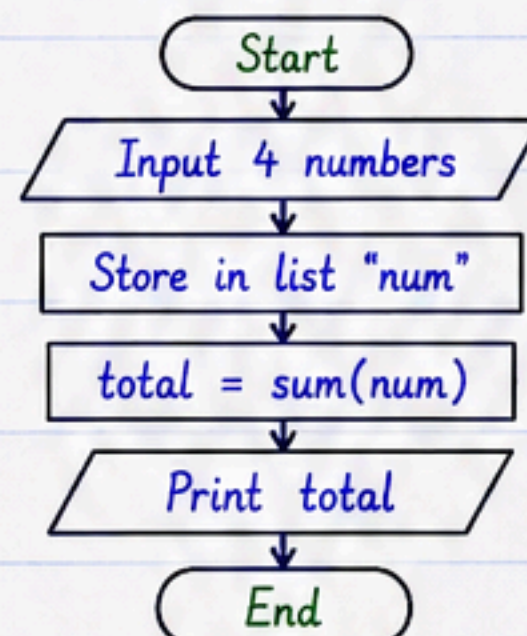
Tuples are immutable, i.e., cannot be changed.

4. Write a program to sum a list with 4 numbers.



Logic / Steps :

1. Take input of 4 numbers.
2. Store in a list.
3. Find the sum using sum().
4. Display the sum.



Example Output

Sum of the list is: 34

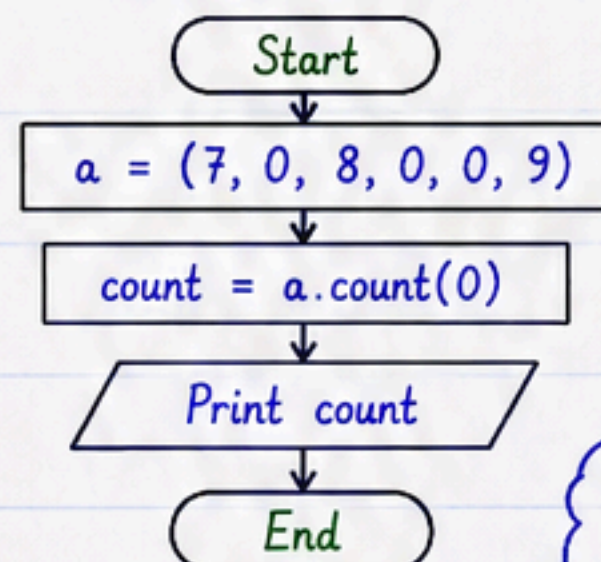
sum() function returns the sum of all items in list.

5. Write a program to count the number of zeros in the following tuple:

a = (7, 0, 8, 0, 0, 9)

Logic / Steps :

1. Take the given tuple.
2. Count how many 0's are there.
3. Display the count.



Example Output

Number of zeros in the tuple: 3

Use count(0) to count zeros in tuple.

